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Tarefa Básica 7

Discussão Sobre Sistemas Lineares - Sistemas Lineares Homogêneos

(Fotos nas páginas seguintes)

Discussão De Sistemas Lineares

Exercício 1:

Tarefa Básica - 7

Tarefa Básica - Discussão de sistema lineares

1-) A-) $\begin{pmatrix} a & 4 & : & 1 \\ 1 & 2 & : & 0,5 \end{pmatrix} \sim \begin{pmatrix} a-2 & 0 & : & 0 \end{pmatrix} \Rightarrow (a-2)x = 0$ falso

-2

$x = \frac{0}{a-2}$

B-) $\Rightarrow x = \frac{0}{a-2} \Rightarrow x = \frac{0}{2-2} \Rightarrow x = \frac{0}{0} \rightarrow$ Indeterminado

Verdadeiro

C-) Se $a \neq 2$, apresenta solução única, ou seja, mais de um valor. falso.

D-) Se $a = 2$, apresenta infinitas soluções falso

$x = \frac{0}{a-2} \Rightarrow \frac{0}{2-2} \Rightarrow \frac{0}{0}$

E-) Como $a = 2$ é indeterminado falso.

Exercício 2:

2-) I-) $\begin{pmatrix} 1 & k & : & 1 \\ k & 1 & : & 1-k \end{pmatrix} \sim \begin{pmatrix} & & & \\ 0 & 1-2k & : & 1-k^2 \end{pmatrix} \Rightarrow y = \frac{1-k^2}{1-2k} \rightarrow$ Não pode ser indeterminado

II-) $k = 0,5$ $y = \frac{1-0,5^2}{1-2 \cdot 0,5} \Rightarrow \frac{0,75}{0} \Rightarrow$ Impossível.

III-) $k \neq 0,5 \Rightarrow$ S.P. Determinado (mais de um valor para k)

$1 - 2k \neq 0$

$1 \neq 2k$

$\boxed{1/2 \neq k}$

$\boxed{\text{D - henrique está correta}}$

Exercício 3 (A e B):

3-) A-)

$$\boxed{A = \begin{pmatrix} 1 & 2 & c \\ 0 & 1 & 1 \\ 3 & 2 & 2 \end{pmatrix}}$$

$$A = \left|\begin{matrix} 1 & 2 & c \\ 0 & 1 & 1 \\ 3 & 2 & 2 \end{matrix}\right| \begin{matrix} 1 & 2 \\ 0 & 1 \\ 3 & 2 \end{matrix}$$

$3c + 2 + 0 = \boxed{3c+2}$

$8 - (3c+2)$

$\boxed{\det A = 6 - 3c}$

$2 + 6 + 0 = \boxed{8}$

B-)

$$\begin{matrix} -2 \\ \\ -2 \end{matrix} \left(\begin{array}{ccc|c} 1 & 2 & c & 1 \\ 0 & 1 & 1 & 2 \\ 3 & 2 & 2 & -1 \end{array}\right) \sim \begin{matrix} \\ \\ -3 \end{matrix} \left(\begin{array}{ccc|c} 1 & 0 & c-2 & -3 \\ & & & \vdots \\ 3 & 0 & 0 & -5 \end{array}\right) \sim \left(\begin{array}{ccc|c} & & & \\ & & & \\ 0 & 0 & -3c+6 & 4 \end{array}\right)$$

$\boxed{\begin{array}{l} \text{S.P. det} \\ D \neq 0 \end{array}}$

$z = \dfrac{4}{-3c+6}$

$-3c + 6 \neq 0$

$6 \neq 3c$

$6/3 \neq c$

$\boxed{c \neq 2}$

$\boxed{\begin{array}{l} \text{c tem que ser} \neq 2 \\ \text{para haver 1 solução.} \\ c \in \mathbb{R} - \{2\} \end{array}}$

Exercício 4:

4-)

$$\begin{matrix} \\ -12 \\ -36 \end{matrix} \left(\begin{array}{ccc|c} 1 & -1 & 0 & k \\ 12 & -k & 1 & 1 \\ 36 & 0 & k & 2 \end{array}\right) \sim \begin{matrix} \\ \\ -k \end{matrix} \left(\begin{array}{ccc|c} 1 & -1 & 0 & k \\ 0 & -k+12 & 1 & 1-12k \\ 0 & 36 & k & 2-36k \end{array}\right) \therefore$$

$$\therefore \left(\begin{array}{ccc|c} & & & \\ & & & \\ 0 & k^2-12k+36 & & 2-12k^2-37k \end{array}\right) \Rightarrow y = \frac{12k^2-37k+2}{k^2-12k+36 \neq 0}$$

$\boxed{k \neq 6}$ → letra E.

$\dfrac{6}{6} + \dfrac{6}{6} = 12$

$\dfrac{6}{6} \cdot \dfrac{6}{6} = 36$

$36 \,|\, 6$

6

Exercício 5 e 6:

5-)

$$\begin{array}{c} \\ -2 \\ -1 \end{array}\begin{pmatrix} 1 & -1 & 1 & \vdots & 6 \\ 2 & 1 & -1 & \vdots & -3 \\ 1 & 2 & -1 & \vdots & -5 \end{pmatrix} \sim \begin{array}{c} \\ \\ -1 \end{array}\begin{pmatrix} 0 & 3 & -3 & \vdots & -15 \\ 0 & 3 & -2 & \vdots & -11 \end{pmatrix} \sim \begin{pmatrix} & & -1 & & \\ 0 & 0 & 1 & \vdots & 4 \end{pmatrix}$$

$\boxed{z=4}$ | $3y-3.4=-15$ | $x+1+4=6$

$3y-12=-15$ | $x+5=6$

$3y=-3$ | $\boxed{x=1}$

$\boxed{y=-1}$

S.P. Determinado

$x.y.z$

$\boxed{1.(-1).4=-4}$ Letra B.

6-)

$$\begin{array}{c} \\ -k \\ -1 \end{array}\begin{pmatrix} 1 & 1 & 1 & \vdots & k \\ k & 1 & 1 & \vdots & 1 \\ 1 & 1 & -1 & \vdots & k \end{pmatrix} \sim \begin{pmatrix} & & & & \\ 0 & 1-k & 1-k & \vdots & 1-k^2 \\ 0 & 0 & -2 & \vdots & 0 \end{pmatrix} \to -2z=0$$

$\boxed{z=\frac{0}{2}}$

$\to z=0$

$(1-k)y+(1+k).0=1-k^2$

$(1-k)y=1-k^2$

$y=\frac{1-k^2}{1-k} \to 1-k^2=0$

$k^2=1$

$1-k=0$

$\boxed{k=1}$

$\boxed{k=1}$

Se $\boxed{k=1} \to$ S.P. Indeterminado

Letra D.

Exercício 7:

7.)

$$\begin{pmatrix} 1 & 1 & 1 = 1 \\ m & -2 & 4 = 5 \\ m^2 & 4 & 16 = 25 \end{pmatrix} \sim \begin{pmatrix} m+2 & 0 & 6 = 7 \\ m^2-4 & 0 & 12 = 21 \end{pmatrix} \sim \begin{pmatrix} & & \\ m^2-2m-8 & 00 & = 7 \end{pmatrix}$$

$$x = \frac{7}{m^2-2m-8}$$

$$-2 + 4 = 2$$

$$-2 \cdot 4 = -8$$

-8 | 2, 4

$x' + x''$

$\boxed{-2 + 4 = 2}$ Letra B.

Tarefa Básica 8

Sistemas Lineares Homogêneos

Exercício 1:

Tarefa Básica 8

Sistemas Lineares Homogêneos

1-) $\begin{pmatrix} x \\ y \end{pmatrix}$

$$\begin{pmatrix} 1 & 7 \\ 7 & 1 \end{pmatrix} \begin{pmatrix} x+7y \\ 7x+y \end{pmatrix} = \begin{pmatrix} xk \\ yk \end{pmatrix} \rightarrow \begin{cases} x+7y = kx \\ 7x+y = ky \end{cases} \therefore$$

$$\therefore \begin{array}{c} -7 \\ \\ \end{array} \left(\begin{array}{cc|c} 1 & 7 & k \\ 7 & 1 & k \end{array}\right) \sim \left(\begin{array}{cc|c} & & \\ 0 & -48 & -6k \end{array}\right) \rightarrow -48y = -6k$$

$$y = \frac{-6k \div 6}{-48 \div 6}$$

$$\boxed{k = 8y} \leftarrow y = \frac{k}{8} \leftarrow y = \frac{-k}{-8} \; (\div -1)$$

$$x + 7y = kx$$

$$x + 7y = 8yx$$

$$\frac{x}{x} = 8y - 7y$$

$$\boxed{1 = y}$$

$$k = 8y$$

$$k = 8 \cdot 1$$

$$\boxed{k = 8} \; m$$

$$\rightarrow \boxed{k = 8} \; m \rightarrow \text{letra E}$$

Exercício 2 e 3:

2-) $\begin{cases} 3x + 4y - z = 0 \\ 2x - y + 3z = 0 \\ x + y + 0 = 0 \end{cases}$ $\left|\begin{array}{ccc:c} 3 & 4 & -1 & 0 \\ 2 & -1 & 3 & 0 \\ 1 & 1 & 0 & 0 \end{array}\right| \sim \left(\begin{array}{ccc:c} 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{array}\right)$ $(\div 11)$

$\therefore \left(\begin{array}{ccc:c} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{array}\right)$ 2 filas iguales $\boxed{\text{Indeterminado (Varias Soluciones)}}$ → Letra D

$1 - x + y = 0$ | $\boxed{y = \alpha}$

$x + \alpha = 0$

$\boxed{x = -\alpha}$

$2x - y + 3z = 0$

$-2\alpha - \alpha + 3z = 0$

$-3\alpha + 3z = 0$

$3z = 3\alpha$

$\boxed{z = \alpha}$

$V = \{(-\alpha, \alpha, \alpha)\}$

$\sim|\sim$

$3 + 4k + 3k = \boxed{3 + 7k}$

3-) $\begin{cases} x + y + z = 0 \\ kx + 3y + 4z = 0 \\ x + ky + 3z = 0 \end{cases}$ $\left|\begin{array}{ccc|cc} 1 & 1 & 1 & 1 & 1 \\ k & 3 & 4 & k & 3 \\ 1 & k & 3 & 1 & k \end{array}\right|$

$13 + k^2 - (3 + 7k)$

$\boxed{k^2 - 7k + 10 = 0}$

$9 + 4 + k^2 = \boxed{13 + k^2}$

$k' + k''$

$2 + 5 = 7$

$\boxed{2 + 5 = 7}$

Letra D

$\begin{cases} \underline{2} + \underline{5} = 7 \\ \underline{2} \cdot \underline{5} = 10 \end{cases}$

$\begin{array}{c|c} 10 & 5 \\ 2 & \end{array}$

Exercício 4 e 5:

4-) $\begin{cases} x + 0 + kz = 0 \\ kx + y + 0 = 0 \\ x + ky + 0 = 0 \end{cases}$ $\begin{pmatrix} 1 & 0 & k & \vdots & 0 \\ k & 1 & 0 & \vdots & 0 \\ 1 & k & 0 & \vdots & 0 \end{pmatrix} \sim \begin{array}{c} \begin{pmatrix} 1 & 0 & k & \vdots & 0 \\ & & & & \\ 1-k^2 & 0 & 0 & \vdots & 0 \end{pmatrix} \\ \begin{array}{ccc} x & y & z \end{array} \end{array}$

$X = \frac{0}{1-k^2 \neq 0} \Rightarrow 1 - k^2 \neq 0$ (solução única $\neq 0$) $\Rightarrow \boxed{k \neq 0}$

$1 \neq k^2$

$\boxed{\pm 1 \neq k}$

$\boxed{V = \{k \in \mathbb{R} / k \neq 0, k \neq 1, k \neq -1\}}$

Letra A.

~||~

5-) $\begin{cases} -x + 2y - 3 = 0 \\ 3x - y + 3 = 0 \\ 2x - 4y + 6 = 0 \end{cases}$ $\begin{pmatrix} -1 & 2 & \vdots & 3 \\ 3 & -1 & \vdots & -3 \\ 2 & -4 & \vdots & -6 \end{pmatrix} \sim \begin{array}{c} \begin{pmatrix} 0 & 5 & \vdots & 6 \\ 0 & 0 & \vdots & 0 \end{pmatrix} \\ \begin{array}{cc} x & y \end{array} \end{array} \Rightarrow 5y = 6$ $\boxed{y = \frac{6}{5}}$

$-x + 2 \cdot \frac{6}{5} = 3 \Rightarrow -x + \frac{12}{5} = 3 \Rightarrow -x = 3 - \frac{12}{5} \therefore$

$\therefore -x = \frac{3}{5} \; (\cdot -1) \Rightarrow \boxed{x = -\frac{3}{5}}$ $\boxed{\text{é determinado}}$

Letra B